

Project Requirement Document

Week 6 Final Capstone | Petalnex Pvt. Ltd. AI Automation Internship

1. Problem Statement

Recruitment teams repeatedly collect applications, extract skills and experience, summarize candidates, compare them against role requirements, request human review, communicate decisions and maintain candidate records. Manual processing creates repetitive work, inconsistent evaluation, delayed responses and avoidable data entry errors.

2. Current Process and Pain Points

- Applications arrive through a form or similar intake channel.
- Recruiters manually inspect candidate information.
- Skills and experience are repeatedly extracted and summarized.
- Candidate evaluation can be inconsistent when performed under time pressure.
- Human decisions and AI recommendations can become mixed together unless the data model separates them.
- Email communication and database updates are repetitive.
- Failures in external services can leave records in ambiguous states.

3. Proposed Solution

- Use an n8n workflow as the orchestration layer.
- Validate application data before any AI or external action.
- Store the candidate record early so every application has traceability.
- Use Gemini for structured extraction, summarization and advisory evaluation.
- Use deterministic rules for role requirements and combine them with the AI score.
- Require human review before the final hiring status is written.
- Trigger interview communication for shortlisted candidates and rejection communication for rejected candidates.
- Log failures and alert an administrator.

4. AI Role

- Extract relevant skills and experience from submitted information.
- Produce a concise candidate summary.
- Provide an advisory score and recommendation.
- Return only a predefined structured schema.
- Never make the final hiring decision.

5. Success Criteria

- Every valid application receives a unique candidate ID.
- Required fields are validated before AI processing.
- AI output is structured and rejected when it violates the schema.
- AI remains advisory and cannot make the final hiring decision.

- Human review is mandatory before shortlist or rejection is finalized.
- Candidate, AI recommendation, human decision and final status remain separate fields.
- Automated emails are sent only after a valid final status exists.
- Failures are logged without exposing credentials or sensitive secrets.
- At least 15 normal, edge, failure and security tests pass.

6. Scope Boundaries

- The system assists recruiters rather than replacing them.
- Evaluation is limited to job relevant evidence.
- Protected characteristics are not used as decision criteria.
- Credentials are stored in n8n credentials or secure environment configuration rather than workflow text.
- Human review is a required control before final status.

Core Candidate Data Model

Field	Type	Purpose
candidate_id	uuid/text	Stable candidate identifier
candidate_name	text	Candidate full name
candidate_email	text	Candidate contact email
phone	text	Optional contact number
skills	text/json	Submitted and extracted skills
experience_years	numeric	Relevant experience
cv_text	text	Normalized CV/profile text
ai_summary	text	AI generated candidate summary
ai_score	numeric	AI advisory score
rule_score	numeric	Deterministic score
combined_score	numeric	60% rule + 40% AI
ai_recommendation	text	shortlist/review/reject
human_decision	text	shortlist/reject
final_status	text	Final business status
human_reviewed_at	timestamp	Review audit timestamp
email_status	text	Email processing status
error_message	text	Failure details without secrets
created_at	timestamp	Record creation time